

Performance Testing

Date	28 Jun 2026
Team ID	XXXXXX
Project Name	OptiCrop
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust / Postman API Testing Suite
Type of Testing	Load Testing and Stress Testing
Target Module / API	/predict endpoint (Crop Recommendation Machine Learning Engine API)
Test Environment	Local Deployment (Flask backend running on local host server environment)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Scenario 1: Smart Crop Recommendation for Farmers - Simultaneous model execution request submission containing simulated N, P, K, temperature, humidity, pH, and rainfall parameters.	50	60	System responds successfully with crop classification predictions under normal load with low response lag.
2	Scenario 2: Crop Suitability Assessment - Concurrent lookups evaluating environmental parameters to check compatibility scores.	30	60	Stable processing with accurate validation return metrics.

3	Scenario 3: Agricultural Research & Policy Queries - Multi-parameter analysis simulation representing continuous policy planning requests.	20	60	System processes long-running analytical requests concurrently without thread exhaustion.
4	Stress Peak Load Simulation - Merging all scenarios simultaneously to test processing limits.	150	120	Graceful handling without server crashes or drop-offs. Max error rate remains below 1%.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.45 seconds	Pass	Fast inference response time from model.
2	Response Time (Max)	< 5 seconds	1.20 seconds	Pass	Observed peak during sudden load step-up.
3	Throughput (Req/sec)	--	42 req/sec	Pass	Highly efficient processing capacity.
4	Error Rate	< 1%	0.00%	Pass	No socket exceptions or timeouts dropped.
5	CPU Utilization	< 80%	45%	Pass	Inference calculations smoothly optimized.
6	Memory Utilization	< 80%	62%	Pass	Flask application context memory footprint stable.

Step 4: Observations & Analysis

Key Findings:

The performance tests prove the OptiCrop engine effectively handles multi-user request execution. Under standard workloads simulating farmers submitting soil metrics (N, P, K, pH, rainfall), the response times safely remained under 0.5 seconds, satisfying target expectations. The Flask-based model serving endpoints successfully ran without memory leaks or model thread locking.

Bottlenecks Identified:

During stress spikes containing up to 150 concurrent virtual users, brief latency climbs occurred as the system processed concurrent matrix inputs for the multi-class model. The CPU utilization grew linearly alongside simultaneous mathematical computations, but did not cause server dropouts.

Optimization Steps Taken:

To maintain stable throughput, model input parsing was modularized to prevent blocking execution pipelines. Array array inputs have been optimized using NumPy operations to minimize raw loop delays before final evaluation by the model.